

5 Key Variables — Pipeline Reference Card

GenAI4Drug · Mahzabeen Project · ChemBERTa-based Molecule Generation Pipeline

1. TEMPERATURE = 1.0		Stage 2 — molecule generation	
What it does:	Divides ChemBERTa logits before softmax. Controls sharpness of token probability distribution at each masked position.		
Why this value:	1.0 = neutral baseline; uses the model's trained probabilities without modification.		
Value to Test	Effect		
0.5	More confident picks. Higher validity, lower diversity. Molecules stay close to original ligand.		
1.0 (current)	Balanced. Neutral sampling from trained distribution.		
1.5	More random. Higher diversity and novelty, but more invalid SMILES generated.		

2. TOP_K = 10		Stage 2 — molecule generation	
What it does:	Limits token sampling to the top K most probable tokens at each masked position. Works together with TEMPERATURE.		
Why this value:	10 restricts generation to chemically plausible tokens while still allowing variety. Avoids sampling nonsensical vocabulary.		
Value to Test	Effect		
3	Very restricted. Conservative molecules, higher validity, sharp drop in diversity.		
10 (current)	Balanced trade-off between validity and chemical diversity.		
50	Wide range of tokens sampled. More novel/unusual molecules, more invalid SMILES.		

3. INCREMENTAL_NUM_SAMPLES = 200		Stage 2 — molecule generation	
What it does:	Number of ChemBERTa generation attempts per (ligand x mask_count x strategy) cell. Total cost scales linearly: 5 ligands x ~8 steps x 2 strategies x 200 = ~16,000 forward passes.		
Why this value:	Code comment: 100 = fast/test, 500 = publication quality. 200 is a practical middle ground.		
Value to Test	Effect		
50	Fast but noisy. Unique SMILES counts unreliable; some cells may show 0 by chance.		
200 (current)	Reasonably reliable diversity estimates. Moderate runtime.		
500	Smooth, reproducible plots. 2.5x more compute. Recommended before publishing results.		

4. RANDOM_MASK_SEED = 47		Stage 1.5 — random mask baseline	
What it does:	Seeds Python random.Random for selecting which atoms are randomly masked in the Stage 1.5 baseline. Ensures reproducibility across runs.		
Why this value:	47 is an arbitrary fixed integer. The value does not matter; fixedness does — it makes the IA vs. random comparison reproducible.		
Value to Test	Effect		
42	Different atom selection. Random curve may shift up/down depending on which atoms are picked.		
47 (current)	Current reproducible baseline.		
Multiple seeds	Best practice: test seeds 42, 47, 100 and confirm IA consistently outperforms random.		

5. INCLUDE_TYPES (list of 8)		Stage 1 — mask calculation	
What it does:	Defines which PLIP protein-ligand interaction types determine which atoms receive a <mask> token. Only atoms involved in listed interaction types are masked.		
Why this value:	All 8 types included = most inclusive setting. Ensures no interaction type is excluded from the biological hypothesis.		
Value to Test	Effect		
"hbond", "saltBridge"	Fewest masked atoms. Fastest pipeline. Misses hydrophobic contacts dominant in BRD4.		
All 8 (current)	Every interacting atom is a candidate. Broadest Stage 2 curves and most informative comparison.		
"hydrophobic", "piStacking", "piCation"	Biologically motivated for BRD4 (hydrophobic/aromatic pocket). Tighter, more targeted redesign.		

TOP_K and TEMPERATURE interact: meaningful diversity increase requires raising both. INCLUDE_TYPES is the only variable encoding biological knowledge — it defines what is redesigned, not how.